



1. Description

1.1. Project

Project Name	STM32
Board Name	custom
Generated with:	STM32CubeMX 6.17.0
Date	06/01/2026

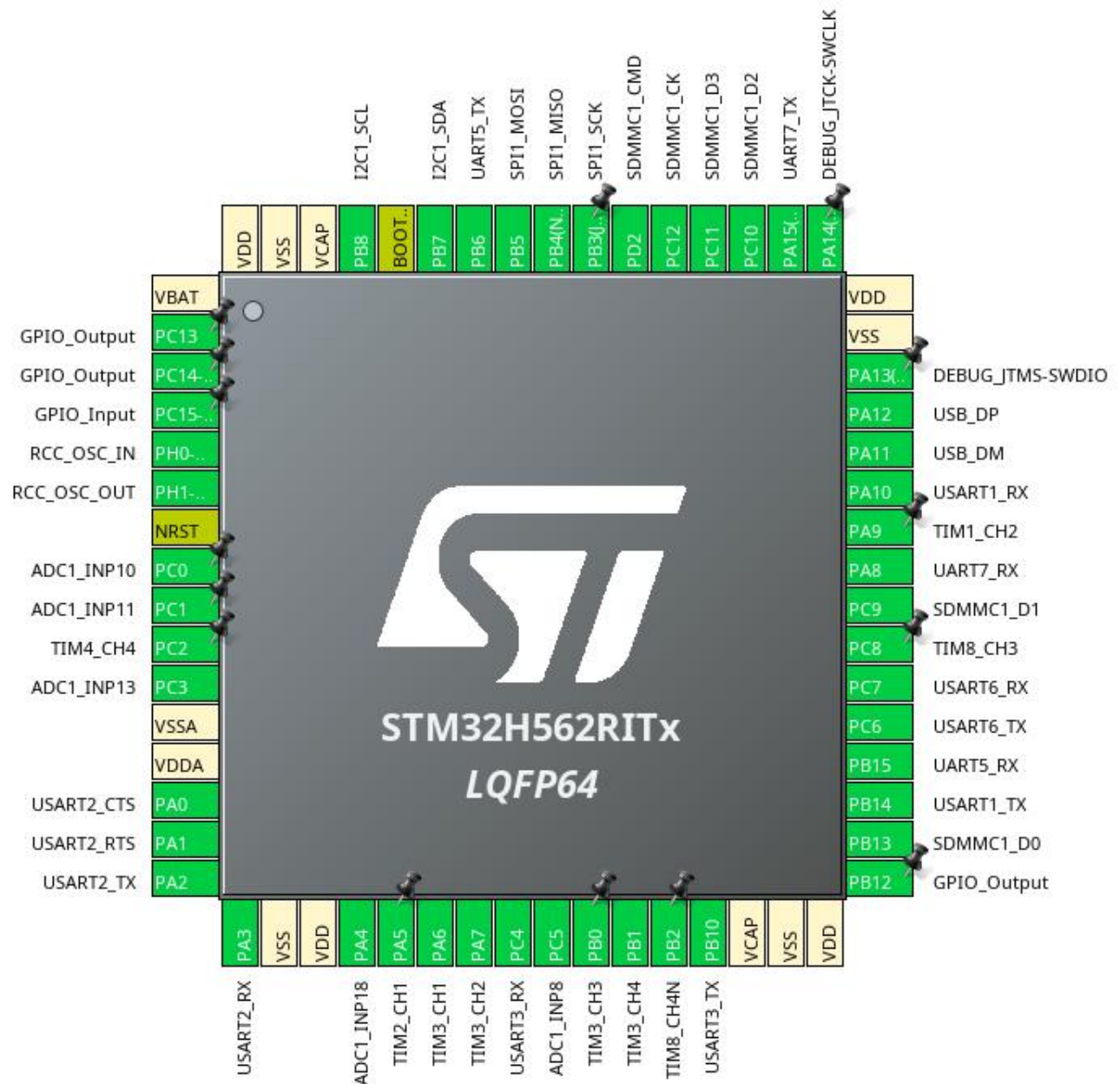
1.2. MCU

MCU Series	STM32H5
MCU Line	STM32H562
MCU name	STM32H562RITx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	ARM Cortex-M33
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2. Pinout Configuration



3. Pins Configuration

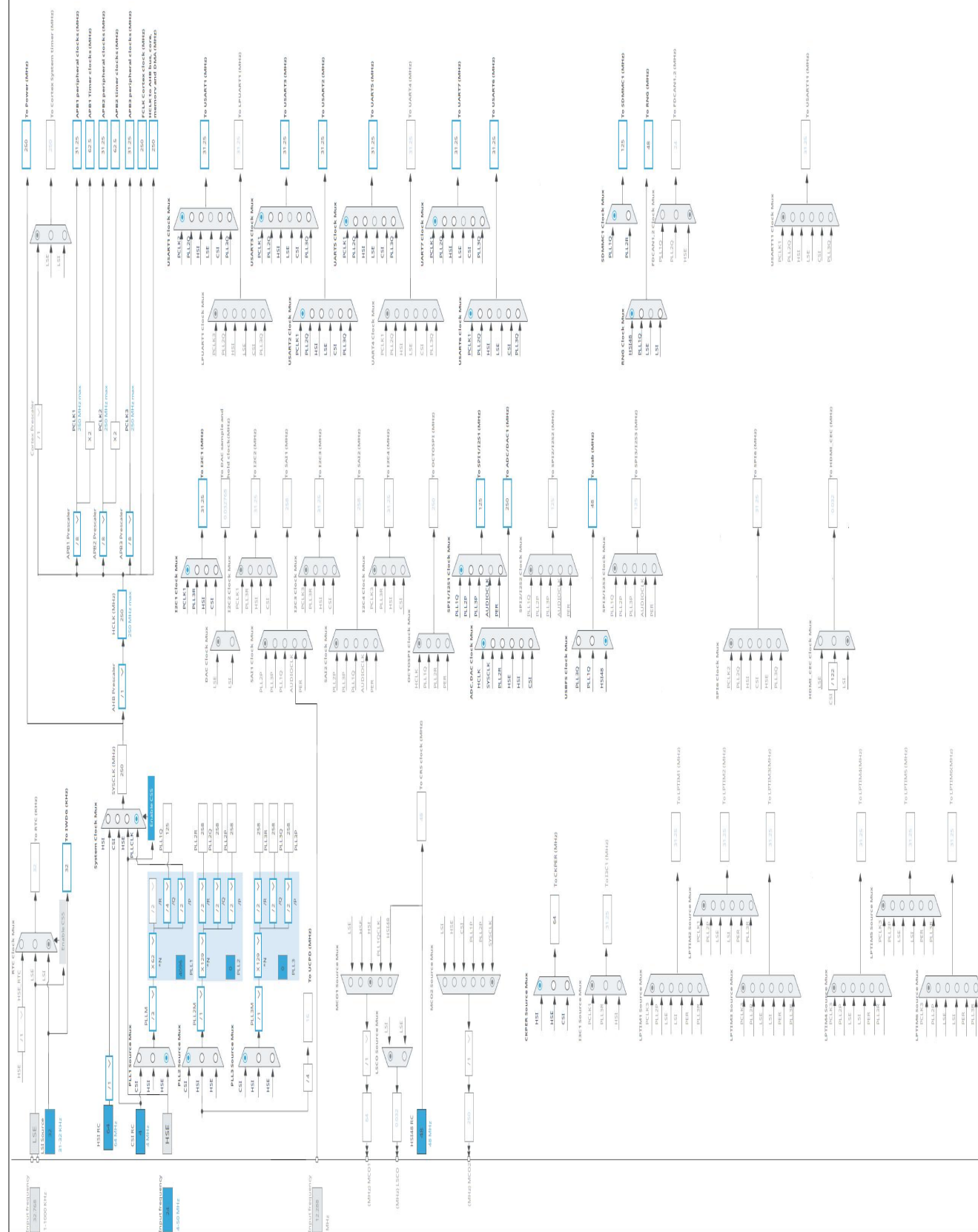
Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
2	PC13 *	I/O	GPIO_Output	
3	PC14- OSC32_IN(OSC32_IN) *	I/O	GPIO_Output	
4	PC15- OSC32_OUT(OSC32_OUT) *	I/O	GPIO_Input	
5	PH0-OSC_IN(PH0)	I/O	RCC_OSC_IN	
6	PH1-OSC_OUT(PH1)	I/O	RCC_OSC_OUT	
7	NRST	Reset		
8	PC0	I/O	ADC1_INP10	
9	PC1	I/O	ADC1_INP11	
10	PC2	I/O	TIM4_CH4	
11	PC3	I/O	ADC1_INP13	
12	VSSA	Power		
13	VDDA	Power		
14	PA0	I/O	USART2_CTS	
15	PA1	I/O	USART2_RTS	
16	PA2	I/O	USART2_TX	
17	PA3	I/O	USART2_RX	
18	VSS	Power		
19	VDD	Power		
20	PA4	I/O	ADC1_INP18	
21	PA5	I/O	TIM2_CH1	
22	PA6	I/O	TIM3_CH1	
23	PA7	I/O	TIM3_CH2	
24	PC4	I/O	USART3_RX	
25	PC5	I/O	ADC1_INP8	
26	PB0	I/O	TIM3_CH3	
27	PB1	I/O	TIM3_CH4	
28	PB2	I/O	TIM8_CH4N	
29	PB10	I/O	USART3_TX	
30	VCAP	Power		
31	VSS	Power		
32	VDD	Power		
33	PB12 *	I/O	GPIO_Output	

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
34	PB13	I/O	SDMMC1_D0	
35	PB14	I/O	USART1_TX	
36	PB15	I/O	UART5_RX	
37	PC6	I/O	USART6_TX	
38	PC7	I/O	USART6_RX	
39	PC8	I/O	TIM8_CH3	
40	PC9	I/O	SDMMC1_D1	
41	PA8	I/O	UART7_RX	
42	PA9	I/O	TIM1_CH2	
43	PA10	I/O	USART1_RX	
44	PA11	I/O	USB_DM	
45	PA12	I/O	USB_DP	
46	PA13(JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
47	VSS	Power		
48	VDD	Power		
49	PA14(JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
50	PA15(JTDI)	I/O	UART7_TX	
51	PC10	I/O	SDMMC1_D2	
52	PC11	I/O	SDMMC1_D3	
53	PC12	I/O	SDMMC1_CK	
54	PD2	I/O	SDMMC1_CMD	
55	PB3(JTDO/TRACESWO)	I/O	SPI1_SCK	
56	PB4(NJTRST)	I/O	SPI1_MISO	
57	PB5	I/O	SPI1_MOSI	
58	PB6	I/O	UART5_TX	
59	PB7	I/O	I2C1_SDA	
60	BOOT0	Boot		
61	PB8	I/O	I2C1_SCL	
62	VCAP	Power		
63	VSS	Power		
64	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration

RIF configuration not available for STM32H562RITx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H5
Line	STM32H562
MCU	STM32H562RITx
Datasheet	DS00000 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

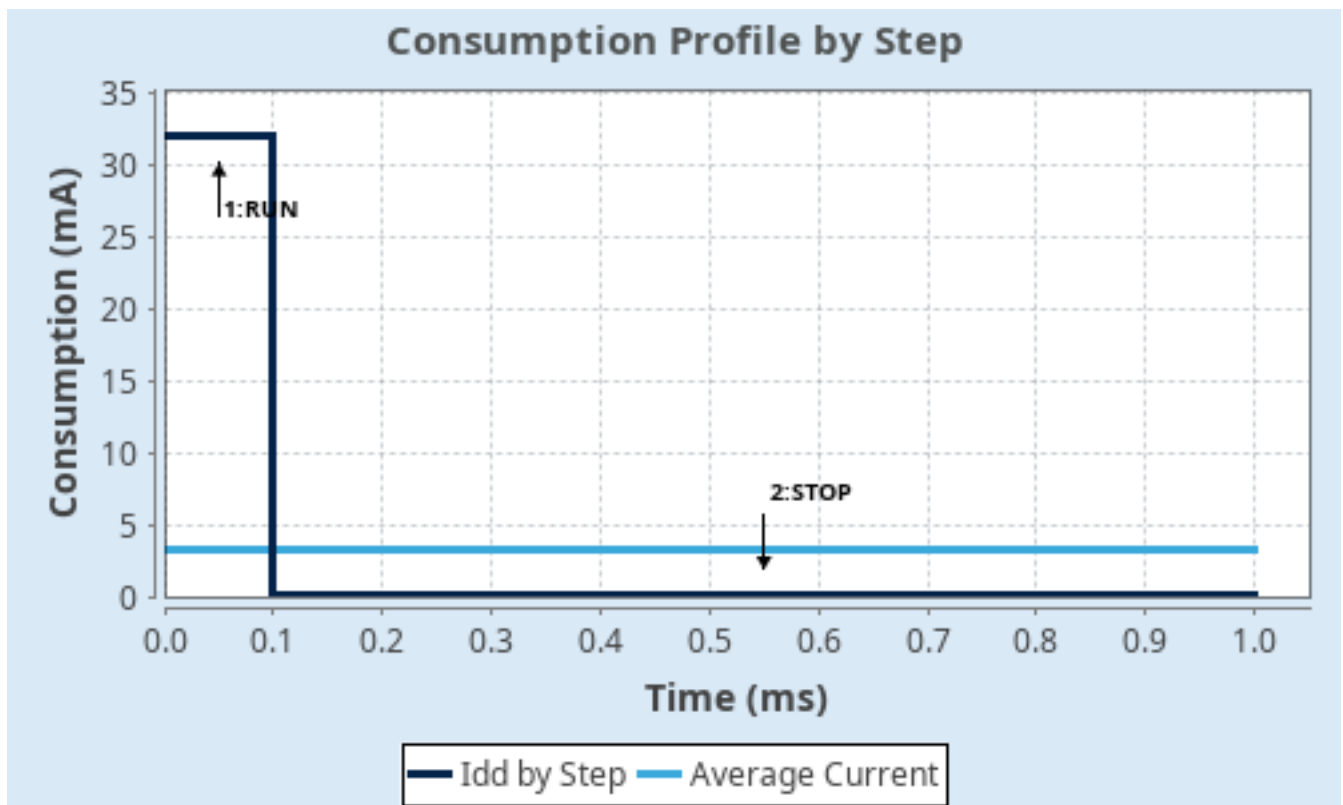
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0	SVOS5: System-Scale5/SMPS
Fetch Type	FLASH_ON/Cache2Ways_A LL_RAM_RETENTION	Flash-PwrDwn_PwrDwnStop_OFF
CPU Frequency	250 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL_CLOCKS_OFF
Clock Source Frequency	8 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	32 mA	51.5 μ A
Duration	0.1 ms	0.9 ms
DMIPS	535.0	0.0
Ta Max	120.68	124.99
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.25 mA
Battery Life	1 month, 13 days, 4 hours	Average DMIPS	535.0 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	STM32
Project Folder	/home/cesil/Desktop/H5 FC/STM32
Toolchain / IDE	EWARM V9.20
Firmware Package Name and Version	STM32Cube FW_H5 V1.6.0
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_GPDMA1_Init	GPDMA1
4	MX_GPDMA2_Init	GPDMA2
5	MX_CORDIC_Init	CORDIC
6	MX_CRC_Init	CRC
7	MX_DCACHE1_Init	DCACHE1
8	MX_FLASH_Init	FLASH
9	MX_FMAC_Init	FMAC
10	MX_I2C1_Init	I2C1
11	MX_IWDG_Init	IWDG

Rank	Function Name	Peripheral Instance Name
12	MX_RAMCFG_Init	RAMCFG
13	MX_RNG_Init	RNG
14	MX_SDMMC1_SDIO_Init	SDMMC1
15	MX_SPI1_Init	SPI1
16	MX_UART5_Init	UART5
17	MX_UART7_Init	UART7
18	MX_USART1_UART_Init	USART1
19	MX_USART2_UART_Init	USART2
20	MX_USB_PCD_Init	USB
21	MX_WWDG_Init	WWDG
22	MX_ADC1_Init	ADC1
23	MX_ICACHE_Init	ICACHE
24	MX_TIM1_Init	TIM1
25	MX_TIM3_Init	TIM3
26	MX_USART6_UART_Init	USART6
27	MX_TIM2_Init	TIM2
28	MX_TIM4_Init	TIM4
29	MX_TIM8_Init	TIM8
30	MX_USART3_UART_Init	USART3

3. Peripherals and Middlewares Configuration

3.1. ADC1

mode: IN8

IN10: IN10 Single-ended

IN11: IN11 Single-ended

mode: IN13

IN18: IN18 Single-ended

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 4
Resolution	ADC 12-bit resolution
Scan Conversion Mode	Disabled
Data Alignment	Right alignment
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
Sampling Mode	Normal
<u>Rank</u>	1
Channel	Channel 8
Sampling Time	2.5 Cycles
Offset Number	No offset
Monitored by	None

ADC_Injected_ConversionMode:

Enable Injected Conversions	Disable
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Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false
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Analog Watchdog 2:

Enable Analog WatchDog2 Mode false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode false

3.2. BOOTPATH

mode: Activated

3.3. CORDIC

mode: Activated

3.4. CORTEX_M33_NS

SysTick Source Selection: HCLK

3.4.1. Parameter Settings:

OTP/RO areas default mode Settings:

Enable OTP/RO area cache disable Enabled *

Cortex Memory Protection Unit Control Settings:

MPU Control Mode Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers

Cortex Memory Protection Unit Region 0 Settings:

Region Enabled
Region Base Address 0x08FFF000 *
Region Limit Address 0x08FFFFFF *
Attributes Number ATTRIBUTE 0
Access Permission ALL READS Permissions
Instruction Access Disable
Shareability Permission Disable

Cortex Memory Protection Unit Region 1 Settings:

Region Disabled

Cortex Memory Protection Unit Region 2 Settings:

Region Disabled

Cortex Memory Protection Unit Region 3 Settings:

Region Disabled

Cortex Memory Protection Unit Region 4 Settings:

Region Disabled

Cortex Memory Protection Unit Region 5 Settings:

Region Disabled

Cortex Memory Protection Unit Region 6 Settings:

Region Disabled

Cortex Memory Protection Unit Region 7 Settings:

Region Disabled

3.4.2. Attributes Settings:

Cortex Memory Protection Unit Attributes 0 Settings:

Attributes Enabled
Memory type Normal non-cacheable

Cortex Memory Protection Unit Attributes 1 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 2 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 3 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 4 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 5 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 6 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 7 Settings:

Attributes Disabled

3.5. CRC

mode: Activated

3.5.1. Parameter Settings:

Basic Parameters:

Default Polynomial State Enable
Default Init Value State Enable

Advanced Parameters:

Input Data Inversion Mode None
Output Data Inversion Mode Disable
Input Data Format Bytes

3.6. DCACHE1

mode: Activated

3.6.1. Parameter Settings:

Basic Parameters:

DCACHE Read Burst Type	DCACHE READ BURST WRAP
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3.7. DEBUG

Debug: Serial Wire

3.8. FLASH

mode: Enable

3.8.1. HAL Option Bytes Settings:

Advanced settings:

Activate	false
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Product State:

Activate	false
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BOR Level:

Activate	false
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User Configuration:

Activate	false
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User Configuration 2:

Activate	false
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Boot Configuration:

Activate NS BOOT LOCK	false
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Write sector group Protection:

Activate Bank 1 WRP	false
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Activate Bank 2 WRP	false
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OTP write protection:

Activate	false
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Flash data sectors:

Activate Bank1 EDATA	false
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Activate Bank2 EDATA	false
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Flash HDP:

Activate Bank1 HDP	false
Activate Bank2 HDP	false

3.8.2. Block-based sector protection:

AREA1:

Activate AREA 1	false
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AREA2:

Activate AREA 2	false
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AREA3:

Activate AREA 3	false
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AREA4:

Activate AREA 4	false
-----------------	-------

AREA5:

Activate AREA 5	false
-----------------	-------

AREA6:

Activate AREA 6	false
-----------------	-------

AREA7:

Activate AREA 7	false
-----------------	-------

AREA8:

Activate AREA 8	false
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3.9. FMAC

mode: Mode

3.10. I2C1

I2C: I2C

3.10.1. Parameter Settings:

Timing configuration:

I2C Speed Mode	Fast Mode *
I2C Speed Frequency (KHz)	400
Rise Time (ns)	100 *
Fall Time (ns)	100 *
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x00600F30 *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.11. ICACHE

Mode: 1-way (direct mapped cache)

3.12. IWDG

mode: Activated

3.12.1. Parameter Settings:

Watchdog Clocking:

IWDG counter clock prescaler	4
IWDG window value	4095
IWDG down-counter reload value	4095

Watchdog Interrupt:

IWDG Early Wakeup Interrupt	0
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3.13. MEMORYMAP

mode: Activated

3.14. PWR

mode: Analog voltage detector

Power Voltage Detector In: Power Voltage Detector In (Internal analog voltage)

mode: Power saving mode

mode: Privilege attributes

3.14.1. Parameter Settings:

Programmable_Voltage_Detector_Settings:

AVD detection Level	PWR AVD LEVEL 0 (1.7 V)
PWR AVD Mode	basic mode is used
PVD detection Level	PWR PVD LEVEL 0 (1.95 V)
PWR PVD Mode	basic mode is used

3.14.2. PWR Privilege :

Privilege PWR:

PWR Privilege	Disable
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3.15. RAMCFG

mode: RAMCFG SRAM1

mode: RAMCFG SRAM2

mode: RAMCFG SRAM3

mode: RAMCFG BKPSRAM

3.15.1. Parameter Settings:

SRAM1:

Hardware Erase	Disable
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SRAM2:

Hardware Erase	Disable
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Error code correction (ECC)	Disable
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Write Protection	Disable
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SRAM3:

Hardware Erase	Disable
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Error code correction (ECC)	Disable
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BKPSRAM:

Hardware Erase	Disable
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Error code correction (ECC)	Disable
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Write Protection	Disable
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3.16. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.16.1. RCC Privilege :

Privilege RCC:

Privilege of RCC Non-Secure Items	Disable
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3.16.2. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	5 WS (6 CPU cycle)
Flash Programming Delay	2

RCC Parameters:

HSI Calibration Value	64
CSI Calibration Value	32
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
TIM Prescaler Selection	Disabled

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 0
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PLL1/2/3 Parameters:

PLL1 input frequency range	Between 8 and 16 MHz
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3.17. RNG

mode: Activated

3.17.1. Parameter Settings:

Clock Error Detection	Enable
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3.18. SDMMC1

Mode: SDIO 4 bits Wide bus

3.18.1. Parameter Settings:

SDMMC parameters:

Clock transition on which the bit capture is made	Rising transition
SDMMC Clock output enable when the bus is idle	Disable the power save for the clock
SDMMC hardware flow control	The hardware control flow is disabled
SDMMC clock divide factor	0
Is external transceiver present ?	no

3.19. SPI1

Mode: Full-Duplex Master

3.19.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits *
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	62.5 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

CRC Parameters:

CRC Calculation	Disabled
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Advanced Parameters:

NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled
Ready Master Management	Internal
Ready Signal Polarity	High

3.20. SYS

Timebase Source: SysTick

3.21. TIM1

Channel2: PWM Generation CH2

3.21.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable

Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.22. TIM2

Clock Source : Internal Clock

Channel1: PWM Generation CH1

3.22.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (32 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.23. TIM3

Clock Source : Internal Clock

Channel1: PWM Generation CH1

Channel2: PWM Generation CH2

Channel3: PWM Generation CH3

Channel4: PWM Generation CH4

3.23.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
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Trigger Event Selection TRGO Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source Disable

PWM Generation Channel 1:

Mode PWM mode 1
Pulse (16 bits value) 0
Output compare preload Enable
Fast Mode Disable
CH Polarity High

PWM Generation Channel 2:

Mode PWM mode 1
Pulse (16 bits value) 0
Output compare preload Enable
Fast Mode Disable
CH Polarity High

PWM Generation Channel 3:

Mode PWM mode 1
Pulse (16 bits value) 0
Output compare preload Enable
Fast Mode Disable
CH Polarity High

PWM Generation Channel 4:

Mode PWM mode 1
Pulse (16 bits value) 0
Output compare preload Enable
Fast Mode Disable
CH Polarity High

3.24. TIM4

Clock Source : Internal Clock

Channel4: PWM Generation CH4

3.24.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value) 0
Counter Mode Up
Dithering Disable
Counter Period (AutoReload Register - 16 bits value) 65535
Internal Clock Division (CKD) No Division

auto-reload preload	Disable
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Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source Disable

PWM Generation Channel 4:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.25. TIM8

Channel3: PWM Generation CH3

Channel4: PWM Generation CH4N

3.25.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High

BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 4N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CHN Polarity	High
CHN Idle State	Reset

3.26. UART5

Mode: Asynchronous

3.26.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable

Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.27. UART7

Mode: Asynchronous

3.27.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.28. USART1

Mode: Asynchronous

3.28.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.29. USART2

Mode: Asynchronous

Hardware Flow Control (RS232): CTS/RTS

3.29.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.30. USART3

Mode: Asynchronous

3.30.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable

Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.31. USART6

Mode: Asynchronous

3.31.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.32. USB

Mode: Device_Only

3.32.1. Parameter Settings:

Basic Parameters:

Speed	Full Speed 12MBit/s
Physical interface	Internal Phy

Signal start of frame	Disabled
-----------------------	----------

Power Parameters:

Low Power	Disabled
-----------	----------

Link Power Management	Disabled
-----------------------	----------

Battery Charging	Disabled
------------------	----------

EndPoint Parameters:

Bulk double buffer	Disabled
--------------------	----------

Iso single buffer	Disabled
-------------------	----------

3.33. WWDG

mode: Activated

3.33.1. Parameter Settings:

Watchdog Clocking:

WWDG counter clock prescaler	1
------------------------------	---

WWDG window value	64
-------------------	----

WWDG free-running downcounter value	64
-------------------------------------	----

Watchdog Interrupt:

Early wakeup interrupt	Disable
------------------------	---------

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC0	ADC1_INP10	Analog mode	No pull-up and no pull-down	n/a	
	PC1	ADC1_INP11	Analog mode	No pull-up and no pull-down	n/a	
	PC3	ADC1_INP13	Analog mode	No pull-up and no pull-down	n/a	
	PA4	ADC1_INP18	Analog mode	No pull-up and no pull-down	n/a	
	PC5	ADC1_INP8	Analog mode	No pull-up and no pull-down	n/a	
DEBUG	PA13(JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
	PA14(JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
I2C1	PB7	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PB8	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
RCC	PH0-OSC_IN(PH0)	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT(PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
SDMMC1	PB13	SDMMC1_D0	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PC9	SDMMC1_D1	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PC10	SDMMC1_D2	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PC11	SDMMC1_D3	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PC12	SDMMC1_CK	Alternate Function Push Pull	No pull-up and no pull-down	High	
	PD2	SDMMC1_CMD	Alternate Function Push Pull	No pull-up and no pull-down	High	
SPI1	PB3(JTDO/TRACESWO)	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB4(NJTRST)	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB5	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM1	PA9	TIM1_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM2	PA5	TIM2_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM3	PA6	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA7	TIM3_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB0	TIM3_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB1	TIM3_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM4	PC2	TIM4_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM8	PB2	TIM8_CH4N	Alternate Function Push Pull	No pull-up and no pull-down	Low	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PC8	TIM8_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
UART5	PB15	UART5_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB6	UART5_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
UART7	PA8	UART7_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA15(JTDI)	UART7_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART1	PB14	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA10	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART2	PA0	USART2_CTS	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA1	USART2_RTS	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA2	USART2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA3	USART2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART3	PC4	USART3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB10	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART6	PC6	USART6_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC7	USART6_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USB	PA11	USB_DM	n/a	n/a	n/a	
	PA12	USB_DP	n/a	n/a	n/a	
GPIO	PC13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC14- OSC32_IN(OSC32_IN)	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC15- OSC32_OU T(OSC32_O	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	
	PB12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	

4.2. GPDMA1

Channel 7 - 8 Words Internal FIFO / 2D addressing: Standard Request Mode

Channel 6 - 8 Words Internal FIFO / 2D addressing: Standard Request Mode

Channel 5 - 8 Words Internal FIFO : Standard Request Mode

Channel 4 - 8 Words Internal FIFO : Standard Request Mode

Channel 3 - 2 Words Internal FIFO : Standard Request Mode

Channel 2 - 2 Words Internal FIFO : Standard Request Mode

Channel 1 - 2 Words Internal FIFO : Standard Request Mode

Channel 0 - 2 Words Internal FIFO : Standard Request Mode

4.2.1. All Channels:

Channel 0:

Request

DMA_REQUEST_SW

Channel 1:

Request DMA_REQUEST_SW

Channel 2:

Request DMA_REQUEST_SW

Channel 3:

Request DMA_REQUEST_SW

Channel 4:

Request DMA_REQUEST_SW

Channel 5:

Request DMA_REQUEST_SW

Channel 6:

Request DMA_REQUEST_SW

Channel 7:

Request DMA_REQUEST_SW

4.2.2. SECURITY:

CH7:

Enable Channel as Privileged NON PRIVILEGED

CH6:

Enable Channel as Privileged NON PRIVILEGED

CH5:

Enable Channel as Privileged NON PRIVILEGED

CH4:

Enable Channel as Privileged NON PRIVILEGED

CH3:

Enable Channel as Privileged NON PRIVILEGED

CH2:

Enable Channel as Privileged NON PRIVILEGED

CH1:

Enable Channel as Privileged NON PRIVILEGED

CH0:

Enable Channel as Privileged NON PRIVILEGED

4.2.3. CH7:

Circular configuration:

Circular Mode Disable

Request Configuration:

Request	SoftWare
DMA Handle in IP Structure	NONE
Block HW request protocol	Single/Burst Level

Channel configuration:

Priority	Low
Transaction Mode	Normal
Direction	Memory To Memory

Source Data Setting:

Source Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Destination Data Setting:

Destination Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Data Handling:

Data Handling Configuration	Disable
-----------------------------	---------

Trigger:

Trigger Configuration	Disable
-----------------------	---------

2D Addressing:

Configuration	Disabled
---------------	----------

Transfer Event Configuration:

Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of each block
---------------------------	---

4.2.4. CH6:

Circular configuration:

Circular Mode	Disable
---------------	---------

Request Configuration:

Request	SoftWare
DMA Handle in IP Structure	NONE
Block HW request protocol	Single/Burst Level

Channel configuration:

Priority	Low
Transaction Mode	Normal
Direction	Memory To Memory

Source Data Setting:

Source Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Destination Data Setting:

Destination Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Data Handling:

Data Handling Configuration	Disable
-----------------------------	---------

Trigger:

Trigger Configuration	Disable
-----------------------	---------

2D Addressing:

Configuration	Disabled
---------------	----------

Transfer Event Configuration:

Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of each block
---------------------------	---

4.2.5. CH5:

Circular configuration:

Circular Mode	Disable
---------------	---------

Request Configuration:

Request	SoftWare
DMA Handle in IP Structure	NONE
Block HW request protocol	Single/Burst Level

Channel configuration:

Priority	Low
Transaction Mode	Normal
Direction	Memory To Memory

Source Data Setting:

Source Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Destination Data Setting:

Destination Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Data Handling:

Data Handling Configuration	Disable
-----------------------------	---------

Trigger:

Trigger Configuration	Disable
-----------------------	---------

Transfer Event Configuration:

Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of each block
---------------------------	---

4.2.6. CH₄:

Circular configuration:

Circular Mode Disable

Request Configuration:

Request SoftWare

DMA Handle in IP Structure	NONE
----------------------------	------

Block HW request protocol	Single/Burst Level
---------------------------	--------------------

Channel configuration:

Priority Low

Transaction Mode Normal

Direction	Memory To Memory
-----------	------------------

Source Data Setting:

Source Address Increment After Transfer **Enabled ***

Data Width	Byte
------------	------

Burst Length	1
--------------	---

Allocated Port for Transfer	Port 0
-----------------------------	--------

Destination Data Setting:

Destination Address Increment After Transfer	Disabled
--	----------

Data Width	Byte
------------	------

Burst Length	1
--------------	---

Allocated Port for Transfer	Port 0
-----------------------------	--------

Data Handling:

Data Handling Configuration	Disable
-----------------------------	---------

Trigger:

Trigger Configuration Disable

Transfer Event Configuration:

Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of each block
---------------------------	---

4.2.7. CH3:

Circular configuration:

Circular Mode Disable

Request Configuration:

Request SoftWare
DMA Handle in IP Structure NONE
Block HW request protocol Single/Burst Level

Channel configuration:

Priority Low
Transaction Mode Normal
Direction Memory To Memory

Source Data Setting:

Source Address Increment After Transfer Disabled
Data Width Byte
Burst Length 1
Allocated Port for Transfer **Port 1 ***

Destination Data Setting:

Destination Address Increment After Transfer Disabled
Data Width Byte
Burst Length 1
Allocated Port for Transfer Port 0

Data Handling:

Data Handling Configuration Disable

Trigger:

Trigger Configuration Disable

Transfer Event Configuration:

Transfer Event Generation The TC (and the HT) event is generated at the (respectively half) end of each block

4.2.8. CH2:

Circular configuration:

Circular Mode Disable

Request Configuration:

Request SoftWare
DMA Handle in IP Structure NONE
Block HW request protocol Single/Burst Level

Channel configuration:

Priority Low
Transaction Mode Normal
Direction Memory To Memory

Source Data Setting:

Source Address Increment After Transfer Disabled

Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Destination Data Setting:

Destination Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Data Handling:

Data Handling Configuration	Disable
-----------------------------	---------

Trigger:

Trigger Configuration	Disable
-----------------------	---------

Transfer Event Configuration:

Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of each block
---------------------------	---

4.2.9. CH1:

Circular configuration:

Circular Mode	Disable
---------------	---------

Request Configuration:

Request	SoftWare
DMA Handle in IP Structure	NONE
Block HW request protocol	Single/Burst Level

Channel configuration:

Priority	Low
Transaction Mode	Normal
Direction	Memory To Memory

Source Data Setting:

Source Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Destination Data Setting:

Destination Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Data Handling:

Data Handling Configuration	Disable
-----------------------------	---------

Trigger:

Trigger Configuration Disable

Transfer Event Configuration:

Transfer Event Generation The TC (and the HT) event is generated at the (respectively half) end of each block

4.2.10. CH0:

Circular configuration:

Circular Mode Disable

Request Configuration:

Request SoftWare
DMA Handle in IP Structure NONE
Block HW request protocol Single/Burst Level

Channel configuration:

Priority Low
Transaction Mode Normal
Direction Memory To Memory

Source Data Setting:

Source Address Increment After Transfer Disabled
Data Width Byte
Burst Length 1
Allocated Port for Transfer Port 0

Destination Data Setting:

Destination Address Increment After Transfer Disabled
Data Width Byte
Burst Length 1
Allocated Port for Transfer Port 0

Data Handling:

Data Handling Configuration Disable

Trigger:

Trigger Configuration Disable

Transfer Event Configuration:

Transfer Event Generation The TC (and the HT) event is generated at the (respectively half) end of each block

4.3. GPDMA2

Channel 7 - 8 Words Internal FIFO / 2D addressing: Linked-List Mode

Channel 6 - 8 Words Internal FIFO / 2D addressing: Linked-List Mode

Channel 5 - 8 Words Internal FIFO : Linked-List Mode

Channel 4 - 8 Words Internal FIFO	: Linked-List Mode
Channel 3 - 2 Words Internal FIFO	: Linked-List Mode
Channel 2 - 2 Words Internal FIFO	: Linked-List Mode
Channel 1 - 2 Words Internal FIFO	: Linked-List Mode
Channel 0 - 2 Words Internal FIFO	: Linked-List Mode

4.3.1. All Channels:

Channel 0:

Request LinkedList Hosting

Channel 1:

Request LinkedList Hosting

Channel 2:

Request LinkedList Hosting

Channel 3:

RequestLinkedList Hosting

Channel 4:

Request LinkedList Hosting

Channel 5:

Request LinkedList Hosting

Channel 6:

Request LinkedList Hosting

Channel 7:

Request LinkedList Hosting

4.3.2. SECURITY:

CH7:

Enable Channel as Privileged	NON PRIVILEGED
------------------------------	----------------

CH6:

Enable Channel as Privileged	NON PRIVILEGED
------------------------------	----------------

CH5:

Enable Channel as Privileged	NON PRIVILEGED
------------------------------	----------------

CH4:

Enable Channel as Privileged	NON PRIVILEGED

CH3:

Enable Channel as Privileged	NON PRIVILEGED

CH₂:

Enable Channel as Privileged	NON PRIVILEGED

CH1:

Enable Channel as Privileged NON PRIVILEGED

CH0:

Enable Channel as Privileged NON PRIVILEGED

4.3.3. CH7:

Channel Configuration for Linked List:

Priority	Low
Execution Mode (circular/linear) of the Linked List	Linear
Linked List Execution Mode	The List is fully executed
Allocated Port for Transfer	Port 0
Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of the last linked-list item

4.3.4. CH6:

Channel Configuration for Linked List:

Priority	Low
Execution Mode (circular/linear) of the Linked List	Linear
Linked List Execution Mode	The List is fully executed
Allocated Port for Transfer	Port 0
Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of the last linked-list item

4.3.5. CH5:

Channel Configuration for Linked List:

Priority	Low
Execution Mode (circular/linear) of the Linked List	Linear
Linked List Execution Mode	The List is fully executed
Allocated Port for Transfer	Port 0
Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of the last linked-list item

4.3.6. CH4:

Channel Configuration for Linked List:

Priority	Low
Execution Mode (circular/linear) of the Linked List	Linear

Linked List Execution Mode	The List is fully executed
Allocated Port for Transfer	Port 0
Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of the last linked-list item

4.3.7. CH3:

Channel Configuration for Linked List:

Priority	Low
Execution Mode (circular/linear) of the Linked List	Linear
Linked List Execution Mode	The List is fully executed
Allocated Port for Transfer	Port 0
Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of the last linked-list item

4.3.8. CH2:

Channel Configuration for Linked List:

Priority	Low
Execution Mode (circular/linear) of the Linked List	Linear
Linked List Execution Mode	The List is fully executed
Allocated Port for Transfer	Port 0
Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of the last linked-list item

4.3.9. CH1:

Channel Configuration for Linked List:

Priority	Low
Execution Mode (circular/linear) of the Linked List	Linear
Linked List Execution Mode	The List is fully executed
Allocated Port for Transfer	Port 0
Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of the last linked-list item

4.3.10. CH0:

Channel Configuration for Linked List:

Priority	Low
Execution Mode (circular/linear) of the Linked List	Linear
Linked List Execution Mode	The List is fully executed

Allocated Port for Transfer
Transfer Event Generation

Port 0
The TC (and the HT) event is generated at the (respectively half) end of the last
linked-list item

4.4. LINKEDLIST

4.5. NVIC configuration

4.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
Window WatchDog interrupt	unused		
PVD/AVD through EXTI Line detection Interrupt	unused		
RAMCFG global interrupt	unused		
Flash non-secure global interrupt	unused		
RCC non-secure global interrupt	unused		
GPDMA1 Channel 0 global interrupt	unused		
GPDMA1 Channel 1 global interrupt	unused		
GPDMA1 Channel 2 global interrupt	unused		
GPDMA1 Channel 3 global interrupt	unused		
GPDMA1 Channel 4 global interrupt	unused		
GPDMA1 Channel 5 global interrupt	unused		
GPDMA1 Channel 6 global interrupt	unused		
GPDMA1 Channel 7 global interrupt	unused		
IWDG global interrupt	unused		
ADC1 global interrupt	unused		
TIM1 Break interrupt	unused		
TIM1 Update interrupt	unused		
TIM1 Trigger and Commutation interrupts	unused		
TIM1 Capture Compare interrupt	unused		
TIM2 global interrupt	unused		
TIM3 global interrupt	unused		
TIM4 global interrupt	unused		
I2C1 Event interrupt	unused		
I2C1 Error interrupt	unused		
SPI1 global interrupt	unused		
USART1 global interrupt	unused		
USART2 global interrupt	unused		
USART3 global interrupt	unused		
UART5 global interrupt	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
TIM8 Break interrupt		unused	
TIM8 Update interrupt		unused	
TIM8 Trigger and Commutation interrupts		unused	
TIM8 Capture Compare interrupt		unused	
USB FS global interrupt		unused	
SDMMC1 global interrupt		unused	
USART6 global interrupt		unused	
GPDMA2 Channel 0 global interrupt		unused	
GPDMA2 Channel 1 global interrupt		unused	
GPDMA2 Channel 2 global interrupt		unused	
GPDMA2 Channel 3 global interrupt		unused	
GPDMA2 Channel 4 global interrupt		unused	
GPDMA2 Channel 5 global interrupt		unused	
GPDMA2 Channel 6 global interrupt		unused	
GPDMA2 Channel 7 global interrupt		unused	
UART7 global interrupt		unused	
FPU global interrupt		unused	
Instruction cache global interrupt		unused	
Data cache global interrupt		unused	
CORDIC global interrupt		unused	
FMAC global interrupt		unused	
RNG global interrupt		unused	

4.5.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Middleware										
System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Utilities	Other
CORTEX_M... ✓	DCACHE1 ✓	ADC1 ✓	TIM1 ✓	I2C1 ✓	RNG ✓	CORDIC ✓	DEBUG ✓	PWR ✓	LINKEDLIST	
FLASH ✓	GPDMA1 ✓	TIM2 ✓	SDMMC1 ✓			CRC ✓				
GPDMA2 ✓	GPIO ✓	TIM3 ✓	SP11 ✓			FMAC ✓				
ICACHE ✓	IWDG ✓	TIM4 ✓	UART5 ✓							
NVIC ✓	RAMCFG ✓	TIM8 ✓	UART7 ✓							
RCC ✓	SYS ✓		USART1 ✓							
WWDG ✓			USART2 ✓							
			USART3 ✓							
			USART6 ✓							
			USB ✓							

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h5-bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h5-ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h5-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h5-series-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-entry-level-graphics.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Brochures	https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h5.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1474-security-bulletin-tn1474stpsirt-information-on-softwarebased--microarchitectural-timing-sidechannel-attacks-on-mcus-with-trustzone-for--armv8m-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf

Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-uart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
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